USN

# RV COLLEGE OF ENGINEERING®

**(An Autonomous Institution affiliated to VTU)**

**VII$^{th}$ Semester B. E. Regular / Supplementary Examinations Jan- 2025**

## DEEP LEARNING (ELECTIVE)

***Time: 03 Hours*** ***Maximum Marks: 100***

***Instructions to candidates:***

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.

### PART-A

| | | | |
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| 1 | 1.1 | State the two main characteristics of knowledge representation. | 02 |
| | 1.2 | Given the following feature map: $\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 1 & 2 \\ 3 & 1 & 2 & 0 \\ 1 & 2 & 3 & 4 \end{bmatrix}$ Apply 2X2 Average pooling with stride=2.Compute the pooled output feature map. | 02 |
| | 1.3 | Compare tiled convolution with locally connected layers in terms of parameter sharing and memory requirements. | 02 |
| | 1.4 | Apply a real-world scenario where a CNN and an RNN can be combined, explaining how this combination benefits the task. | 01 |
| | 1.5 | A researcher wants to classify musical notes using a CNN. The audio waveform is first converted into a spectrogram where rows represent frequencies and columns represent time. What type of data format is this (1-D, 2-D, or 3-D), and why is a convolutional model suitable for this representation? | 02 |
| | 1.6 | Explain how mapping data to a lower-dimensional space helps in improving model generalization. | 01 |
| | 1.7 | Given $4 \times 4$ feature map: $\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 1 & 2 \\ 0 & 2 & 3 & 1 \\ 4 & 1 & 2 & 0 \end{bmatrix}$ Apply a $2 \times 2$ max pooling operation with a stride of 2 and determine the resulting output feature map. | 02 |
| | 1.8 | A neural network outputs the values [2, 1, 0] for a 3-class classification problem. Apply the softmax activation function and compute the probability for each class. | 02 |
| | 1.9 | Identify which pooling operation is applied in the transition layer to reduce the feature map size? | 01 |
| | 1.10 | Explain how the transition layer in DenseNet reduces computational complexity and memory usage. | 02 |
| | 1.11 | A model trained to recognize animals is adapted to identify a newly discovered animal species. In one case, only one labeled image of the new species is provided; in another case, no labeled images are available and only semantic descriptions are used. Identify which case corresponds to one-shot learning and which corresponds to zero-shot learning, and justify your answer. | 02 |
| | 1.12 | Identify the layer used after the last dense block to convert feature maps into a feature vector. | 01 |

## PART-B

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| 2 | a | A neuron has three inputs: $x_1 = 0.5, x_2 = -1.0, x_3 = 2.0$ with the corresponding weights $w_1 = 0.8, w_2 = -0.4, w_3 = 0.3$, and bias $b = 0.2$.<br>Compute the output of the neuron for the following activation functions:<br>i) Threshold (Heaviside) function<br>ii) Sigmoid function<br>iii) Hyperbolic Tangent Function<br>Show all intermediate steps, including the calculation of the induced local field, and final outputs. | 08 |
| | b | Compare max pooling, average pooling, *and* L2 pooling, and analyze how each affects the learned feature representations and invariance properties of the network. | 08 |
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| 3 | a | With a neat diagram, briefly explain the different classes of neural network architectures | 08 |
| | b | A 3 × 3 image patch I and a 2 × 2 kernel K are given below:<br>$I = \begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 1 \\ 2 & 3 & 2 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$<br>a) Perform a valid 2-D convolution (no padding, stride = 1) and compute the full 2 × 2 output feature map.<br>b) If the same kernel were applied as a cross-correlation, show which entries would differ.<br>c) State one practical reason why kernel flipping is rarely used in CNN implementations. | 08 |
| | | **OR** | |
| 4 | a | In a CNN, fully shared filters limit spatial flexibility, while fully local filters increase parameter count. Apply an intermediate strategy that balances these two extremes and explain how it reduces memory requirements. | 08 |
| | b | Illustrate how convolution and pooling in convolutional neural networks can be interpreted as imposing an infinitely strong prior on the model parameters. | 08 |
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| 5 | a | A company's shallow RNN model fails to capture long-term dependencies in complex sequential data. How can deep recurrent network architectures improve performance in this situation? Discuss with reference to deeper transformations, hierarchical hidden layers, and skip connections. | 08 |
| | b | A practitioner observes that a recurrent neural network fails to retain past input information over time. Applying the principles of Echo State Networks, describe how adjusting the spectral radius of the recurrent weight matrix can address this issue | 08 |
| | | **OR** | |
| 6 | a | Explain the Encoder–Decoder Sequence-to-Sequence RNN architecture, describing how the encoder forms the context vector and how the decoder generates the output sequence. Also discuss the limitations of using a fixed-size context vector and how attention addresses this issue. | 08 |
| | b | With a neat diagram, elaborate the working principle of LSTM with all relevant equations. | 08 |
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| 7 | a | Infer how autoencoders learn low-dimensional data manifolds by balancing reconstruction and regularization, and how this helps capture on-manifold variations while suppressing off-manifold noise. | 08 |
| | b | Elaborate on the training procedure of a denoising autoencoder, explaining the role of corrupted inputs, the encoder–decoder architecture, and the reconstruction loss. | 04 |
| | c. | Describe the two competing objectives involved in training an autoencoder and explain why balancing reconstruction accuracy with regularization is important. | 04 |
| | | **OR** | |
| 8 | a | Describe Regularized Autoencoders and explain how sparsity, denoising, and contractive penalties help autoencoders learn meaningful representations. | 10 |
| | b | Characterize how autoencoders learn low-dimensional data manifolds by balancing reconstruction and regularization. | 06 |
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| 9 | a | Explain the architecture working principle and properties of a Deep Boltzmann Machine. | 10 |
| | b | Summarize the architecture, key features, and significance of VGGNet and ResNet. | 06 |
| | | **OR** | |
| 10 | a | Explain Batch Normalization and how it improves the training stability and convergence of deep neural networks by normalizing mini-batch activations and using learnable scale and shift parameters. | 10 |
| | b | Write the Early Stopping meta-algorithm used to determine the optimal number of training iterations in deep learning. Clearly indicate the roles of validation error, patience parameter, and storage of best model parameters. | 06 |